

Notes: Nucera, Sarno, and Zinna (RFS, 2024)

Currency Risk Premiums Redux

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1 Big picture

- **Question.** What is the *right* asset-pricing kernel for currency returns, and which macro-financial nontradable factors truly earn currency risk premia once one corrects for omitted-factor and measurement-error biases?
- **Setting.** The paper studies a large cross-section of currency portfolios built from the main currency investment strategies, and then prices those portfolios with latent factors extracted from the return panel.
- **Core challenge.** Standard two-pass Fama–MacBeth procedures are especially fragile for currency risk premia because many candidate factors are nontradable, noisy, and possibly correlated with omitted sources of systematic risk. In FX, that problem is severe precisely because the literature contains many proposed “macro,” “financial,” and “text-based” factors.
- **Main methodological move.** The paper combines the Giglio–Xiu three-pass procedure with Lettau–Pelger risk-premium PCA (RP-PCA). This **augmented three-pass method** uses the cross-section of currency portfolios to recover latent pricing factors and then projects each candidate factor onto those latent factors.
- **Main factor-structure result.** The optimal currency stochastic discount factor (SDF) contains at least **three latent pricing factors**: a strong *dollar* factor plus two weak but high-Sharpe-ratio slope factors that resemble *carry* and *short-term momentum*. Evidence for a fourth *value*-type factor is weaker.
- **Main economic result.** Once the SDF is specified correctly, only a **small fraction** of more than 100 candidate nontradable factors have statistically significant risk premia, and those are mostly related to **volatility, uncertainty, and liquidity** rather than to standard macro variables.
- **Main comparative-statics result.** The standard two-pass method materially overstates the number and size of priced nontradable factors. Accounting for omitted variables and measurement error sharply shrinks the FX “factor zoo.”
- **Why this paper matters.** The paper is really about **how to do robust currency asset pricing**. It shows that correctly identifying the latent pricing kernel comes first, and only then can we talk seriously about the macro-financial sources of currency premia.

2 Data and measurement (key objects)

2.1 FX data and individual currency returns

- **Sample.** Monthly data from October 1983 to December 2017.
- **Currencies.** The underlying universe contains **49 currencies** against the U.S. dollar, including both developed and emerging-market currencies.
- **Investor perspective.** Returns are defined from the point of view of a U.S. investor.
- **Raw objects.** Spot exchange rates and one-month forward rates, both quoted relative to the U.S. dollar.

- **Currency excess return.** For currency i ,

$$X_{i,t+1} = \frac{S_{i,t+1} - F_{i,t}}{S_{i,t}},$$

where $S_{i,t}$ is the spot rate and $F_{i,t}$ is the matching one-month forward rate.

- **Interpretation.** This is the excess return to buying foreign currency forward at t and unwinding in the spot market at $t + 1$.
- **Forward premium.** The forward premium is approximately the interest rate differential relative to the U.S. dollar under covered interest parity, so the same object connects the FX asset-pricing setup to standard carry-trade logic.

2.2 Test assets: currency strategy portfolios

- **Why portfolios rather than individual currencies?** The three-pass procedure requires a reasonably large cross-section of test assets, and portfolios average out currency-specific idiosyncratic noise.
- **Number of test assets.** The baseline panel contains **46 currency portfolios**.
- **Strategies.** The portfolios come from nine well-known tradable strategies:
 - carry,
 - short-term momentum,
 - long-term momentum,
 - net foreign assets (NFA),
 - liabilities in domestic currency (LDC),
 - value,
 - term spread,
 - long-term yields,
 - output gap.
- **Sorting design.** Most strategies sort currencies into five portfolios from low-signal to high-signal currencies. The high-minus-low spread portfolio (HML) summarizes the strategy payoff.
- **Economic meaning.** Moving from $P1$ to $P5$ should move from relatively safer/low-premium currencies toward relatively riskier/high-premium currencies if the strategy truly captures priced risk.

2.3 Candidate nontradable factors

- **Goal.** Estimate the price of risk of nontradable factors after filtering out omitted-variable and measurement-error biases.
- **Number of factors.** The paper studies **134 candidate factors**.
- **Grouping.**

- 24 financial factors,
 - 30 text-based factors,
 - 80 macro factors.
- **Examples of financial factors.** Global FX volatility, equity and bond volatility, liquidity indicators, funding and intermediary-capital measures, and broader financial conditions indicators.
 - **Examples of text-based factors.** News-based economic policy uncertainty (EPU) and equity-market-volatility (EMV) indicators, including topic-specific subindexes.
 - **Examples of macro factors.** Output, inflation, unemployment, consumption, and industrial production variables, often in world or world-minus-U.S. form.
 - **Transformation.** The factors are turned into innovations, in the baseline via AR(1) residuals, and then standardized.
 - **Important feature.** The third pass is run *separately* for each candidate factor, so the choice of one candidate factor does not change the latent SDF or the estimates for the others.

2.4 Unobservables and timing

- **Latent pricing factors.** The true factors that price currency portfolios are unobserved and must be estimated from the panel of portfolio returns.
- **Measurement error.** Candidate nontradable factors are noisy proxies for underlying risks.
- **Omitted variables.** Standard methods may leave out weak but important latent pricing factors, especially those with low time-series variance but high Sharpe ratios.
- **Weak factors.** A factor can matter greatly for pricing even if it explains only limited time-series variation. This is central for the paper’s momentum result.

3 Models and empirical specifications (all equations + interpretation)

3.1 Latent factor representation of test-asset returns

The starting point is a latent factor model for excess returns:

$$X_{nt} = F_t \psi_n' + \varepsilon_{nt}, \quad n = 1, \dots, N, \quad t = 1, \dots, T.$$

Here, X_{nt} is the excess return on test asset n , F_t is the vector of latent pricing factors, ψ_n is the vector of factor loadings, and ε_{nt} is the idiosyncratic residual.

- **Interpretation.** Currency portfolios are driven by a small number of latent systematic risks, plus portfolio-specific noise.
- **What is being learned?** Both the number of relevant latent factors and their economic content.

3.2 Risk-premium PCA (RP-PCA)

Standard PCA uses only return covariances. RP-PCA instead also uses information in expected returns. The key object is

$$\Sigma_{RP} = \frac{1}{T} X'X + \omega \bar{X}'\bar{X},$$

where X is the $T \times N$ matrix of test-asset returns, $\bar{X}$ is the vector of mean returns, and ω is the risk-premium weight.

In population, this matrix converges to

$$\Psi(\Sigma_F + (1 + \omega)\mu_F'\mu_F)\Psi' + \text{Var}(\varepsilon),$$

so the means of the factors matter directly for factor extraction.

RP-PCA can also be written as minimizing a joint objective:

$$\min_{\Psi, F, \bar{F}} \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T (X_{nt} - F_t \psi'_n)^2 + \frac{\omega}{N} \sum_{n=1}^N (\bar{X}_n - \bar{F} \psi'_n)^2.$$

- **Interpretation.** The first term fits the time series of returns; the second term fits the cross-section of average returns.
- **Economic meaning of ω .**
 - $\omega = -1$ collapses RP-PCA to standard PCA,
 - $\omega = 0$ uses second moments rather than covariances,
 - $\omega > 0$ overweights expected returns and helps detect weak but high-Sharpe-ratio pricing factors.
- **Why this matters in FX.** A factor like momentum may be weak in variance terms but still crucial for pricing. Standard PCA can therefore miss it.

3.3 Choosing the currency SDF

Once factors are extracted, the paper evaluates candidate SDFs using the tangency-portfolio representation. If μ_F and Σ_F are the factor means and covariance matrix, the mean-variance tangency weights are

$$\hat{b}_{MV} = \mu_F \Sigma_F^{-1}.$$

The implied SDF is

$$\phi_t = 1 - (\hat{F}_t - \mu_F) \hat{b}'_{MV}.$$

The paper then evaluates models using:

- the maximal Sharpe ratio of the factor span,
- pricing errors,
- idiosyncratic variance,
- statistical tests for the number of factors.

An OLS pricing regression is used to summarize fit:

$$X_{nt} = \alpha_n + \hat{F}_t \psi'_n + \varepsilon_{nt}.$$

From this, one can study pricing-error summaries such as root-mean-square alpha and the magnitude of the idiosyncratic component.

- **Interpretation.** A good SDF should span average returns well, keep pricing errors small, and avoid unnecessarily fitting noise.
- **Core trade-off.** Adding factors can reduce pricing errors, but one wants factors that are economically meaningful and not just statistical leftovers.

3.4 The augmented three-pass procedure

The paper uses the Giglio–Xiu three-pass logic, amended by RP-PCA factor extraction.

First pass: estimate test-asset exposures. After the RP-PCA transformation of returns and factors, the loadings are recovered from

$$\tilde{X}_{nt} = \tilde{F}_t \psi'_n + \varepsilon_{nt}.$$

Second pass: estimate latent factor prices of risk. Cross-sectional average returns are regressed on the estimated exposures:

$$\bar{X}_n = \hat{\psi}_n \gamma' + a_n,$$

where γ is the vector of latent factor prices of risk.

Third pass: price a candidate nontradable factor. Let g_{jt}^t be the innovation in candidate factor j . The factor is projected onto the latent-factor space:

$$g_{jt}^t = a_{jk} + \hat{F}_{1:k,t} \eta'_j + u_{jkt}.$$

The estimated price of risk of the candidate factor is then

$$\hat{\lambda}_{g,j} = \hat{\gamma} \hat{\eta}'_j,$$

and the denoised return-based version of the factor is

$$\hat{g}_{jt} = \hat{F}_t \hat{\eta}'_j.$$

- **Interpretation.** Instead of directly pricing a noisy candidate factor, the method first asks: what part of this factor lies in the space of true currency pricing risks?
- **Why it solves the main problem.** The candidate factor price of risk is computed using the latent pricing space extracted from the test assets, so one does not need to guess the full true SDF in advance.

3.5 Rotation invariance

The key object is the candidate factor price of risk, not the individual latent-factor coefficients by themselves. If the latent factors are rotated by any full-rank matrix H , then

$$\hat{\lambda}_g = \hat{\gamma}\hat{\eta}' = \hat{\gamma}H^{-1}H\hat{\eta}'.$$

So $\hat{\lambda}_g$ is invariant even though the factor loadings and prices of risk are not individually invariant.

- **Interpretation.** The exact statistical representation of the latent factors is not uniquely pinned down, but the economically relevant price of risk of a candidate factor is.

3.6 Interpreting candidate-factor exposures and prices of risk

The projection coefficients η tell us how a candidate factor loads on the latent pricing factors, while λ_g tells us whether the candidate factor is priced.

- **Positive price of risk.** The factor is procyclical: it tends to rise in good states and thus earns a positive premium.
- **Negative price of risk.** The factor is countercyclical: it pays off in bad states and therefore commands a negative premium.
- **Important subtlety.** A candidate factor can load positively on one latent factor and negatively on another. Therefore, the sign of the final risk premium, not just the loadings, determines the economic interpretation.

3.7 Weak candidate factors and SPCA

The paper then checks robustness using supervised principal component analysis (SPCA), which is designed to handle omitted-variable bias, measurement error, and the possibility that the candidate factor is weak in the cross-section.

- **Interpretation.** Even after accounting for weak factors explicitly, the main message remains: financial and text-based factors matter more than standard macro variables.
- **Practical use.** SPCA further filters the list of significant factors by requiring positive out-of-sample hedging power.

3.8 Simulation design

To evaluate finite-sample performance, the paper calibrates a Monte Carlo data-generating process to the FX data. The simulated DGP uses four denoised currency factors: dollar, carry, short-term momentum, and value. The simulations match the empirical dimensions of the baseline sample ($N = 46$, $T = 410$).

- **Why this matters.** Good asymptotics in equity applications do not automatically carry over to FX, where the cross-section is smaller and the factor structure is different.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Omitted-variable bias.** If the researcher leaves out relevant pricing factors from the SDF, first-pass betas and second-pass prices of risk become biased.
- **Measurement-error bias.** If a candidate factor is noisy, the Fama–MacBeth procedure inflates or distorts its estimated price of risk.
- **Why this is especially bad here.** Most macro-financial candidate factors in FX are nontradable, so their means do not identify premia directly and measurement error is often severe.

4.2 What identifies the latent pricing kernel

- **Cross-sectional information.** The 46 portfolio returns across nine strategies provide the key information needed to recover the latent pricing space.
- **Balanced portfolio panel.** Even though the individual currency data are unbalanced, the strategy portfolios are dynamically constructed so the test-asset panel is balanced.
- **Economic variation.** The strategies are designed to load on distinct sources of FX risk, which helps reveal the factor structure of the SDF.

4.3 Why RP-PCA helps identify weak pricing factors

- Standard PCA ranks factors only by variance explained, so it can miss a factor with low time-series variance but high Sharpe ratio.
- RP-PCA uses both covariances and mean returns, so it can recover weak but important pricing factors.
- In the paper, this is exactly what happens with the latent *momentum* factor.

4.4 Why candidate-factor prices of risk are robustly identified

- The candidate factor is not priced in isolation.
- Instead, the method first projects the candidate factor onto the latent pricing space, and only then computes the implied premium.
- This is what makes the estimate robust to both omitted factors and measurement error.
- Rotation invariance ensures that the final candidate-factor premium is economically meaningful even if the individual latent factors are only identified up to rotation.

4.5 Why the additional checks are informative

- **SPCA check.** Shows the core results are not artifacts of weak-factor problems.

- **Recursive and out-of-sample checks.** Show the three-factor structure is fairly stable over time, while the fourth factor is fragile and contributes little out of sample.
- **Monte Carlo analysis.** Shows the augmented three-pass estimator performs well in finite samples tailored to the FX setting.

4.6 Relation to the literature

- Relative to the FX literature, the paper moves from strategy-by-strategy pricing toward a large cross-section asset-pricing framework.
- Relative to the standard Fama–MacBeth approach, it explicitly addresses omitted-factor and measurement-error biases.
- Relative to Giglio–Xiu, it adapts the three-pass logic to FX and improves factor extraction with RP-PCA.
- Relative to the FX “factor zoo,” it provides a disciplined way to decide which candidate factors actually survive once the SDF is correctly specified.

5 Tables and figures

5.1 Figure 1: Currency investment strategies

Read:

- The raw portfolios already show a clear risk-return trade-off: high-signal portfolios tend to have higher volatility and higher average returns.
- The strongest HML strategies are **carry**, **short-term momentum**, and **output gap**, with annualized returns around 7.3%, 6.9%, and 6.7%, respectively.
- Long-term yield and term-spread strategies are much weaker.
- The cumulative-return plots also show that these strategies are not the same risk packaged differently: for example, short-term momentum performs well when carry crashes during the global financial crisis.

5.2 Figure 2 and Table 1: How many latent factors matter?

Read:

- RP-PCA reveals three clear pricing factors and a possible fourth weak factor.
- The first factor is strong in the time-series sense; the second and third are weaker in variance terms but strong in pricing terms because they have high Sharpe ratios.
- A three-factor RP-PCA model substantially improves pricing relative to PCA; the paper reports a maximal Sharpe ratio around **1.57** for the three-factor RP-PCA SDF versus about **0.90** for the corresponding PCA model.

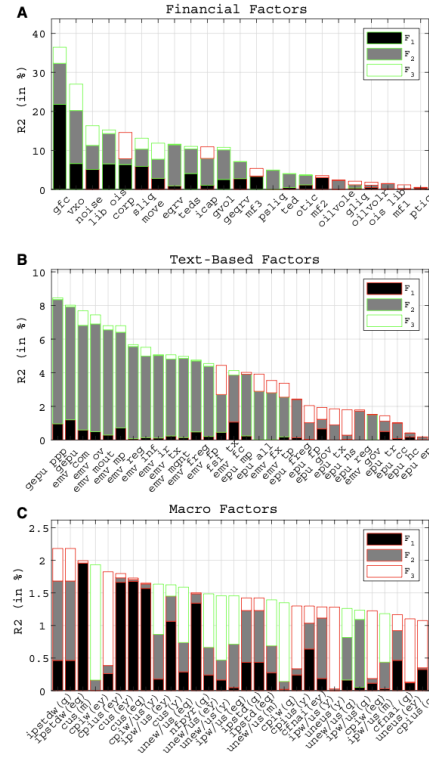


Figure 5
Nontradable factors' explained variation by latent pricing factors
The figure shows nontradable candidate factors' explained variation by latent factors (extracted using RP-PCA with baseline RP-weight $\omega=20$ applied to the panel of $N=46$ currency portfolios). We limit the evidence to the first three extracted factors, given that the optimal SDF is $\varphi(F_{1-k})$ with $k=3$ (Figure A4 in the Internet Appendix uses $k=6$). Panels A, B, and C present the evidence for financial, text-based and macro candidate factors, respectively. In each panel, candidate factors are sorted by the spanning R^2 s associated with the k -factor SDF (to help visualize the results, we only display the 30 macro factors with the highest R^2 s). Bars quantify the latent factor contribution to the overall R^2 . We consider SDFs of expanding dimension, thus k ranges from 1 to 3. Nontradable factors are expressed as innovations, using the residuals from AR(1) processes, and are then standardized. The sample period varies with the factor at hand, according to data availability over the November 1983 to December 2017 period ($T=410$). See factor descriptions in Tables A5–A7 in the Internet Appendix.

- The second factor looks like **carry**, while the third looks like **short-term momentum**. The fourth looks like a **value**-type factor, or a “long value, short long-term momentum” factor.
- The three-factor RP-PCA model prices the cross-section of portfolio returns very well. The two-factor model misses important returns, especially those tied to momentum.
- Under the three-factor model only one portfolio still has a significant average pricing error; under the four-factor model none do.

5.4 Figure 5 and Table 2: How nontradable factors load on the latent pricing kernel

Read:

- Most candidate factors have low spanning R^2 , which is direct evidence that **measurement error is pervasive**.
- Financial factors sometimes have meaningful R^2 's above 10%, and a few very strong cases above 30%. Text-based factors have lower but still informative R^2 's. Macro factors typically have very low R^2 's, often around 1–2%.

- The **carry** latent factor is the main channel through which many financial and text-based factors matter.
- For several financial factors, the loadings on carry and momentum have **opposite signs**. This is a central reason why omitting momentum distorts estimated risk premia.

Table 2
Exposures of nontradable factors to latent factors

	A.I. Risk Exposures		A. Financial Factors						A.II. Explained Variation					
	η_{F1}	η_{F2}	η_{F3}	η_{F4}	η_{F5}	η_{F6}	η_{F7}	η_{F8}	R^2_{F1-2}	R^2_{F1-3}	R^2_{F1-4}	R^2_{F1-5}	R^2_{F1-6}	R^2_{F1-8}
otic	0.73**	5.73***	-0.35	-1.42	2.26	7.47**	1.17	3.67	3.68	3.77	4.01	5.99		
noise	-1.36*	-11.34***	1.34**	8.06*	-0.62	-0.41	4.99	10.98	17.33	20.05	20.07	20.07		
sluq	-1.29**	-10.48***	5.28	7.79*	-0.86	-0.67*	5.81	11.47	13.40	16.21	16.23	17.60		
glic	3.13***	11.71***	-7.49***	-2.96	8.80***	3.13	21.83	32.33	36.43	36.83	40.49	40.84		
gvol	-1.07**	-9.85***	2.99	7.80**	-2.07	-3.34***	2.55	10.13	10.79	13.54	13.74	16.21		
pliq	0.03	8.00**	-0.45	-0.88	2.42	-0.44	0.00	4.91	4.96	5.00	5.27	5.30		
ind	-0.45	-4.82*	-0.44	0.36	0.72	0.27	0.47	3.94	3.58	3.96	3.98	3.98		
lib.iss	-1.48*	-14.14*	3.61	1.30	6.90	4.52	6.39	16.80	16.97	17.09	18.03	18.35		
more	-0.88**	-10.89***	3.81	0.12	-0.25	-1.94	2.78	8.33	11.75	11.75	11.75	11.86		
vno	-1.60***	-15.19***	1.85	-6.91***	-1.35	6.47	19.96	26.40	26.55	28.50	28.50	28.56		
capr	-0.65	-11.72**	1.46	3.11	-2.83	0.80	0.93	11.45	11.6	12.04	12.42	12.44		
gprp	-1.11***	-7.55*	-0.72	4.39*	-0.99	4.82	2.76	7.13	7.17	7.86	8.01	8.84		

	B.I. Risk Exposures		B. Text-Based Factors						B.II. Explained Variation					
	η_{F1}	η_{F2}	η_{F3}	η_{F4}	η_{F5}	η_{F6}	η_{F7}	η_{F8}	R^2_{F1-2}	R^2_{F1-3}	R^2_{F1-4}	R^2_{F1-5}	R^2_{F1-6}	R^2_{F1-8}
gprp	0.00	-11.18***	3.38	3.12	-3.48	-3.52	1.12	7.36	7.64	7.99	8.26	8.56		
gprp.ppp	0.13	-12.01***	3.82	2.81	-3.79	-2.97	0.89	7.82	8.17	8.45	8.75	8.97		
env.rr	-0.48*	-9.13***	2.88	1.87	-1.51	0.31	0.48	6.73	7.35	7.51	7.63	7.61		
env.mout	-0.38	-9.09**	2.19	2.74	-0.31	1.32	0.29	6.40	6.73	7.07	7.07	7.14		
env.mout	-0.36	-7.85**	1.35	2.55	-1.12	1.09	0.14	4.78	4.92	5.21	5.27	5.31		
env.inf	-0.28	-7.93**	2.92	0.09	-0.98	0.29	0.14	4.85	5.45	5.45	5.50	5.50		
env.rr	-0.15	-4.96*	3.68	1.21	-1.47	0.20	0.58	4.60	7.56	7.63	7.79	7.78		
env.rr	-0.23	-7.97**	0.89	1.32	-1.56	1.60	0.11	4.96	5.03	5.11	5.22	5.31		
env.fc	-0.72	-6.25*	2.52	2.78	2.96	-2.17	1.06	3.80	4.16	4.50	4.40	5.06		
env.fv	-0.32	-7.32**	1.52	1.15	-1.58	0.36	0.21	4.29	4.48	4.54	4.65	4.65		
env.v	-0.34	-7.72*	2.09	2.10	-0.68	0.61	0.23	4.68	4.99	5.19	5.21	5.22		
env.rsp	-0.59	-8.40**	2.80	2.88	0.22	1.18	0.70	6.22	6.74	7.11	7.11	7.16		
env.rsp	-0.20	-8.56**	1.84	2.19	1.49	0.19	0.07	5.42	5.63	5.84	5.93	5.93		
env.rsp	-0.40	-7.54*	1.32	2.18	1.18	2.37	0.47	4.60	4.70	4.91	4.96	5.16		

(Continued)

Table 2
(Continued)

	C.I. Risk Exposures					C. Macro Factors					C.II. Explained Variation				
	η_{F1}	η_{F2}	η_{F3}	η_{F4}	η_{F5}	η_{F6}	η_{F7}	η_{F8}	η_{F9}	η_{F10}	R^2_{F1-2}	R^2_{F1-3}	R^2_{F1-4}	R^2_{F1-5}	R^2_{F1-6}
env(y)	-0.49	-2.24	-1.54	-0.92	-0.08	1.49	1.06	1.45	1.62	1.66	1.66	1.66	1.74		
ipw(sic)	0.25	-2.60	-1.80	0.53	0.85	1.87	0.12	0.63	0.87	0.88	0.92	1.05			
ipw(sic)	-0.15	-3.69**	-1.40	1.39	-0.88	1.99	0.03	1.09	1.24	1.32	1.36	1.50			
ipw(sic)	0.10	-1.25	-4.07***	0.51	-0.17	-2.67	0.02	0.14	1.35	1.36	1.36	1.61			
ipw(sic)	-0.36	-1.77	-2.58	-1.97	2.08	-2.59	0.28	0.52	1.01	1.18	1.39	1.63			
news(sic)	-0.27	-1.99	-3.66**	-2.53	-0.64	1.37	0.16	0.47	1.46	1.75	1.77	1.83			

The table presents the nontradable candidate factors' exposures to the latent factors (η_{F1}) and the explained variation (R^2_{F1-1}) obtained from the spanning regressions of Equation (15). For SDFs including an increasing number of factors, $k=1, \dots, K$. We report the candidate factor exposures to the first six extracted, orthogonalized latent factors ($K=6$). The factors are extracted by applying RP-PCA with baseline weight ($\omega=20$) to the $N=46$ portfolios obtained from the nine investment strategies. Panels A, B, and C show the estimates for the financial, text-based, and macro factors, respectively. These factors are extracted as interventions, using the residuals from AR(1) processes, and are then standardized. We present the estimates only for the candidate factors with statistically significant risk premiums ($\hat{\eta}_{F1}$) according to the three-pass model with optimal number of factors, that is, η_{F1-3} , reported later in Table 3. When a macro factor's risk premium is significant at multiple frequencies, we present only the most representative one (the frequency at which the factor's risk premium is most precisely estimated). We report all candidate factors' exposures in Table A10, in the Internet Appendix. See nontradable factor descriptions in Tables A5-A7, in the Internet Appendix. * $p < .1$; ** $p < .05$; *** $p < .01$ denote significance at the 1, 5 and 10% levels, respectively, based on Newey-West standard errors.

5.5 Figure 6 and Table 3: Which nontradable factors are actually priced?

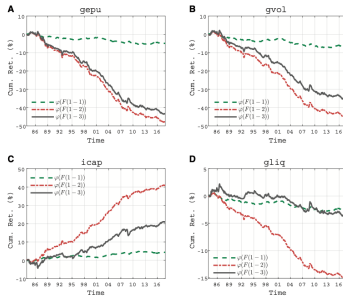


Figure 6
Returns-based candidate factors
The figure shows the cumulative returns of selected return-based candidate factors. Factor A is the global economic policy uncertainty index (gepu); factor B is the global FX volatility factor (gvol); factor C is the financial intermediaries' capital ratio factor (icap); and factor D is the global FX liquidity factor (gliq). For each candidate factor, we present three versions of return-based factors. By expanding the dimension of the SDF, that is, η_{F1-1} with $k=1, 2, 3$. Specifically, for η_{F1-1} , the return-based factor is given by $F_{1,t} \eta_{F1-1}$, where the latent factors are extracted using RP-PCA with $\omega=20$ and the exposures are obtained by estimating the spanning regressions of Equation (15); hence, the underlying exposures are those displayed in the first three columns of Table 2 (for icap and gliq, see Table A10 in the Internet Appendix, as their premium estimates are not statistically significant using the three-factor model). The spanning regression sample period can vary with the factor at hand, according to data availability over the November 1983 to December 2017 period. See the factor descriptions in Tables A5-A7 in the Internet Appendix.

Factor	Risk premium of nontradable factors									
	η_{F1}	η_{F2}	η_{F3}	η_{F4}	η_{F5}	η_{F6}	η_{F7}	η_{F8}	η_{F9}	η_{F10}
gepu	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
gvol	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
icap	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
gliq	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.mout	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.inf	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.fc	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.fv	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.v	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Factor	Risk premium of nontradable factors									
	η_{F1}	η_{F2}	η_{F3}	η_{F4}	η_{F5}	η_{F6}	η_{F7}	η_{F8}	η_{F9}	η_{F10}
gepu	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
gvol	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
icap	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
gliq	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.mout	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.inf	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rr	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.fc	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.fv	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.v	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
env.rsp	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Read:

- The three-pass method sharply reduces the number of significant nontradable factors relative to Fama–MacBeth. The factor zoo shrinks because much of it is noise.
- Among financial factors, the paper highlights the importance of **global FX volatility**, **systematic FX liquidity**, several U.S. bond and equity volatility/liquidity indicators, and some broader financial-conditions measures.
- Among text-based factors, the **global EPU** index stands out. Several EMV measures are also significant, especially those related to macro and monetary-policy uncertainty.
- Macro variables look much weaker. Only a handful survive, often at specific frequencies, and their signs are sometimes hard to reconcile with theory. The broader message is a robust disconnect between standard macro variables and currency returns.

5.6 Table 4, Figure 7, and Table 5: Simulation evidence

Table 4
Simulation calibration and true values

Panel A: $z_{it}^j = \hat{F}_i^j \hat{\xi}_i^T + \epsilon_{it}$					Panel C: $g_{it}^j = \hat{Z}_i^j \hat{\eta}_i^T + u_{jt}$				
A.I: Risk Premiums ($\hat{\gamma}$)					C.I: Risk Premiums ($\hat{\lambda}_Z$)				
$\hat{\gamma}_{F1}$	$\hat{\gamma}_{F2}$	$\hat{\gamma}_{F3}$	$\hat{\gamma}_{F4}$		$\hat{\lambda}_{Zdol}$	$\hat{\lambda}_{Zcar}$	$\hat{\lambda}_{Zmom}$	$\hat{\lambda}_{Zval}$	
0.187	0.113	0.092	0.037	$\hat{\lambda}_Z$	0.026	0.064	0.082	0.022	
A.II: Risk Exposures ($\hat{\xi}$)					C.II: Risk Exposures ($\hat{\eta}_Z$)				
$\hat{\xi}_1$	$\hat{\xi}_2$	$\hat{\xi}_3$	$\hat{\xi}_4$		$\hat{\eta}_{Zdol}$	$\hat{\eta}_{Zcar}$	$\hat{\eta}_{Zmom}$	$\hat{\eta}_{Zval}$	
0.137	0.017	-0.018	-0.014	g_{ivol}^d	-5.275	-12.403	1.900	5.307	
0.000	0.730	0.196	0.080	g_{icap}^d	4.265	11.128	-0.004	6.127	
-0.004	-0.249	0.661	-0.141	g_{ipus}^d	-2.435	1.665	1.932	0.885	
0.021	0.026	0.074	0.762		C.III: Signal-to-Noise ($g; \hat{\sigma}; \hat{u}$)				
A.III: Signal-to-Noise ($z; \hat{\sigma}; \hat{\xi}$)					σ_u	$\sigma_{\hat{g}}$	$\sigma_{\hat{g}}^2/\sigma_u^2$	$\sigma_{\hat{g}}^2/\sigma_{\hat{g}}^2$	
σ_z	σ_z	σ_z^2/σ_z^2	σ_z^2/σ_z^2	g_{ivol}^d	11.172	4.421	0.135	0.157	
0.019	0.246	175.218	0.994	g_{icap}^d	11.288	4.089	0.116	0.131	
0.135	0.258	3.667	0.786	g_{ipus}^d	11.980	1.024	0.007	0.007	
0.213	0.228	1.140	0.533		Panel D: True Risk Premiums				
0.160	0.203	1.621	0.618		$\hat{\lambda}_Z$	$\hat{\lambda}_G$	(se)	SR	
Panel B: $X_{nt} = \hat{Z}_t \hat{\psi}_t^T + \epsilon_{nt}$					z_{dol}	0.026	—	0.013	0.367
B.I: Risk Exposures ($\hat{\psi}_Z$)					z_{car}	0.064	—	0.014	0.854
$\hat{\psi}_{Zdol}$	$\hat{\psi}_{Zcar}$	$\hat{\psi}_{Zmom}$	$\hat{\psi}_{Zval}$	z_{mom}	0.082	—	0.012	1.249	
mean	1.071	-0.014	-0.008	-0.027	z_{val}	0.022	—	0.010	0.369
med	1.071	-0.065	0.018	-0.012	g_{ivol}^d	—	-0.759	0.360	0.595
min	0.664	-0.465	-0.533	-0.517	g_{icap}^d	—	0.756	0.430	0.641
max	1.359	0.585	0.449	0.498	g_{ipus}^d	—	0.159	0.385	0.539

The table reports all the inputs required in the calibration of the data-generating process. Panel A: $z_{it}^j = \hat{F}_i^j \hat{\xi}_i^T + \epsilon_{it}$ reports: A.I the risk premiums of the latent factors ($\hat{\gamma}$); A.II the risk exposures of the noisy demeaned z -factors to the demeaned latent factors ($\hat{\xi}$); and A.III the standard deviations of the noise in the z -factors (σ_z), the standard deviations of the denoised z -factors (σ_z), the signal-to-noise ratios (σ_z^2/σ_z^2), and the explained variations or spanning R^2 s (σ_z^2/σ_z^2). Panel B: $X_{nt} = \hat{Z}_t \hat{\psi}_t^T + \epsilon_{nt}$ reports the average, median, minimum, and maximum risk exposures of the N test assets to each of the four denoised z -factors ($\hat{\psi}_Z$). Panel C: $g_{it}^j = \hat{Z}_i^j \hat{\eta}_i^T + u_{jt}$ reports: C.I the risk premiums of the denoised z -factors ($\hat{\lambda}_Z$); C.II the risk exposures of the noisy candidate nontradable g -factors to the denoised, demeaned z -factors ($\hat{\eta}_Z$); and C.III the standard deviations of the noise in the g -factors (σ_u), the standard deviations of the denoised g -factors ($\sigma_{\hat{g}}$), the signal-to-noise ratios ($\sigma_{\hat{g}}^2/\sigma_u^2$), and the explained variations or spanning R^2 s ($\sigma_{\hat{g}}^2/\sigma_{\hat{g}}^2$). Panel D: True Risk Premiums reports the target risk-premium parameters for the z - and g -factors: for each factor, the risk-premium estimate ($\hat{\lambda}$), the associated asymptotic standard error (se), and the annualized Sharpe ratio (SR). The z -factors are the four tradable factors (dollar, carry, ST mom, and value), and the g -factors are the standardized AR(1) innovations to the selected nontradable factors (gvol, icap, and ipus). To obtain the denoised factors that drive the data-generating process, we applied the baseline three-pass model with $\omega=20$ and four latent factors. Based on this calibration, we then perform $M=10,000$ artificial Monte Carlo realizations with $N=46$ and $T=410$, which match the cross-section and time-series dimensions of our baseline sample of FX portfolio returns.

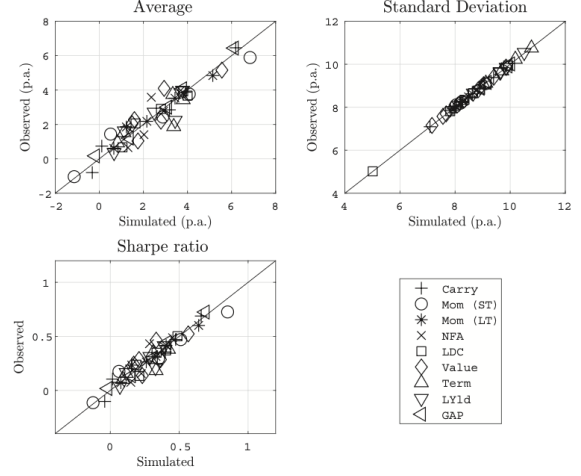


Figure 7
Observed versus simulated test-asset return moments

The figure shows the moments of the observed and simulated test-asset returns. In *Average*, we plot the time-series averages of the observed portfolio returns (*Observed*, in percent annualized) against the means of the time-series averages of the simulated portfolio returns (*Simulated*, in percent annualized); in *Standard deviation*, we show the standard deviations of the observed returns against the means of the standard deviations of the simulated returns (in percent, annualized); and in *Sharpe ratio*, we report the annualized Sharpe ratios of the observed returns against the means of the Sharpe ratios of the simulated returns. The observed cross-sectional standard deviation of the average returns is 0.0173, and the mean of the cross-sectional standard deviation of the simulated average returns is 0.0194. The true data-generating process (DGP) has four factors, and the parameters are calibrated based on the four denoised z -factors (dollar, carry, ST mom, and value). To denoise the z -factors, we use RP-PCA with $\omega=20$ and four pricing factors. We simulate the models 10,000 times with $N=46$ and $T=410$ (see Section VI and Table A24 in the Internet Appendix).

Read:

- The simulated returns match the observed return moments quite closely, so the Monte Carlo exercise is well calibrated to the FX setting.
- The augmented three-pass estimator recovers the true factor risk premia, standard errors, and Sharpe ratios well in finite samples.
- The standard two-pass estimator performs poorly whenever the SDF is misspecified or factors are noisy, which is exactly the empirical environment the paper studies.
- The simulation therefore validates the paper's empirical message: omitted-variable and measurement-error problems are not abstract concerns; they are quantitatively first order in FX pricing.

5.7 Additional robustness and filtering results

Read:

- Out-of-sample and recursive exercises continue to support a **three-factor** SDF. The fourth factor contributes little out of sample.
- SPCA confirms the baseline results and filters the set of significant candidate factors even further.

Table 5
Three-pass simulation results

	A.I. Estimate		A.II. SD		A. Two-Factor Model, $\varphi(F_{1:t-1})$				A.IV. RMSE	
	λ	SE	λ	SE	λ	SE	λ	SE	λ	SE
λ_{dollar}	0.026	0.012	0.367	0.013	0.001	0.170	0.000	-0.001	0.001	0.170
λ_{carry}	0.089	0.012	1.409	0.017	0.001	0.234	0.025	-0.003	0.555	0.031
$\lambda_{\text{momentum}}$	0.013	0.007	1.091	0.024	0.002	0.795	-0.050	-0.005	-0.159	0.055
λ_{vol}	0.007	0.006	0.791	0.010	0.001	0.418	-0.015	-0.004	0.421	0.018
λ_{noise}	-1.356	0.296	1.314	0.343	0.034	0.226	-0.597	-0.064	0.720	0.689
λ_{noise}	1.194	0.286	1.342	0.379	0.033	0.245	0.438	-0.144	0.701	0.579
λ_{noise}	0.189	0.252	0.826	0.258	0.040	0.466	0.030	-0.133	0.288	0.260

(Continued)

Table 5
(Continued)

	C.I. Estimate		C.II. SD		C. Four-Factor Model, $\varphi(F_{1:t-1})$				C.IV. RMSE	
	λ	SE	λ	SE	λ	SE	λ	SE	λ	SE
λ_{dollar}	0.026	0.012	0.368	0.012	0.001	0.169	0.000	-0.001	0.001	0.169
λ_{carry}	0.064	0.013	0.864	0.013	0.001	0.182	0.000	-0.001	0.010	0.013
$\lambda_{\text{momentum}}$	0.082	0.012	1.294	0.012	0.001	0.185	0.000	-0.001	0.085	0.012
λ_{vol}	0.014	0.010	0.360	0.015	0.001	0.221	-0.007	0.000	-0.010	0.017
λ_{noise}	-0.843	0.353	0.672	0.376	0.035	0.289	-0.084	-0.007	0.078	0.385
λ_{noise}	0.724	0.348	0.608	0.356	0.035	0.276	-0.033	-0.082	-0.033	0.357
λ_{noise}	0.178	0.306	0.680	0.307	0.036	0.445	0.019	-0.079	0.141	0.307

The table presents the three-pass estimates on simulated data, whereby the estimator uses baseline RP-weights ($w=20$) and $k=2, \dots, 3$ latent factors ($\varphi(F_{1:t-1})$). Thus, panels A, B, C, and D refer to the two-, three-, four- and five-factor models, respectively. For a selected model, we focus on the following three objects: the candidate factor's annualized risk premium (λ), the associated asymptotic standard error (SE), and the annualized Sharpe ratio of the factor denominated counterpart (SR). Specifically, in each panel, Tab I reports the means of the estimates (λ , SE , and SR), Tab II the standard deviations of the estimates (SD), Tab III the means of the biases (Bias), whereby the bias is given by the estimate minus the true value for the selected object, and Tab IV the root-mean-square errors (RMSE). The true data-generating process (DGP) has four factors, and the parameters are calibrated based on the four denominated λ -factors (dollar, carry, ST mom, and vol). To remove noise from the factors, we applied the three-pass model with $w=20$ and four latent factors ($\varphi(F_{1:t-1})$) to the panel of FX portfolio returns. The candidate factors include the four noisy λ -factors (dollar, carry, ST mom, and vol) and the innovations to the three noisy nontradable λ -factors (λ_{noise} , λ_{noise} , and λ_{noise}). The true values are reported in Table 4, along with the other calibrated parameters. The results are based on $M=10,000$ artificial Monte Carlo realizations with $N=40$ and $T=410$.

- Starting from 134 candidate factors, the number of significant premia is over 90 under standard Fama–MacBeth, falls to **43** under the three-pass method, and falls further to about **18** once SPCA also requires positive out-of-sample fit.

6 Short summary

- **Conclusion.** Currency returns are priced by a small latent factor structure, not by a sprawling list of observable candidate factors.
- **Main factor insight.** The optimal currency SDF has at least three factors: dollar, carry, and momentum. The momentum factor is weak in time-series terms but essential for pricing.
- **Main empirical lesson.** Once omitted-variable and measurement-error biases are addressed, only a small set of mostly financial and text-based factors carry significant currency risk premia.
- **Method lesson.** Robust pricing of nontradable factors requires getting the latent SDF right first.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that robust inference on currency risk premia requires a latent-factor asset-pricing approach that explicitly corrects for omitted-variable and measurement-error biases.
- It shows that the currency pricing kernel is richer than a simple dollar-carry model and must include a momentum-type factor as well.
- It also shows that many proposed nontradable FX factors do not survive once the SDF is properly specified.

7.2 Economic intuition

- Currency strategies may look distinct in raw returns, but many of them are driven by a few common latent risks.
- The dollar factor captures broad common movement across currencies.

- Carry and momentum are slope factors: they explain why long-short strategies earn premia in the cross-section.
- Many observable candidate factors are only noisy reflections of those latent risks. Once the noise is stripped out and omitted factors are restored, only a narrow set of volatility, uncertainty, and liquidity measures remains priced.

7.3 Takeaway

This paper is best read as a methodological and empirical cleanup of modern FX asset pricing. Its deepest message is that the question “Which macro-financial variables price currencies?” cannot be answered with a naive two-pass regression on a hand-picked factor. One must first recover the latent pricing kernel from a large cross-section of currency returns. After that correction, the FX factor zoo looks much smaller: the main pricing action runs through latent dollar, carry, and momentum risks, and the observable nontradable factors that survive are mostly volatility, uncertainty, and liquidity proxies rather than standard macro variables.